

Banking Management System

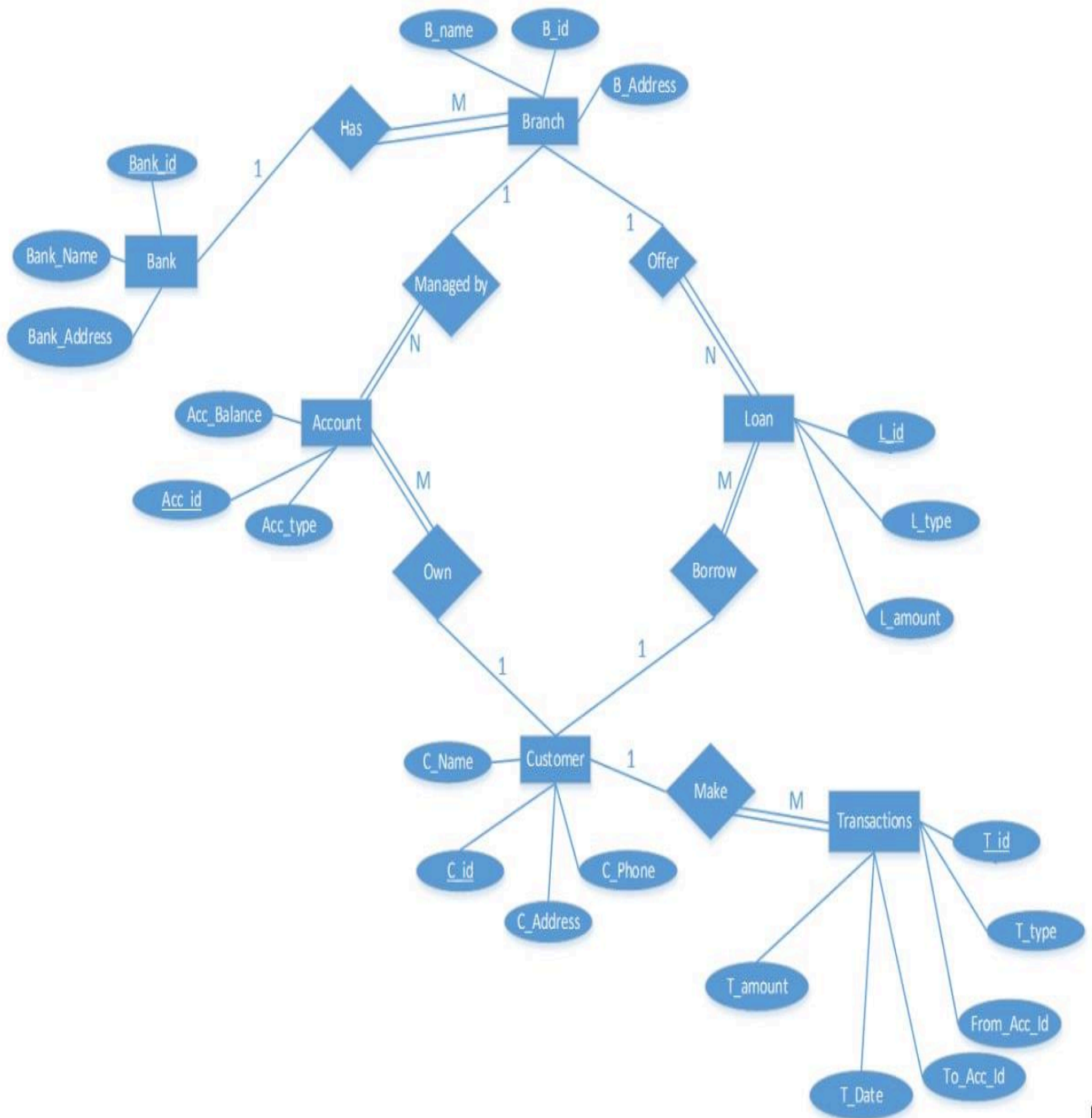
The main processes of Bank is to lend money to coustomer and companies, transfer money between accounts of coustomer or companies, invest money of coustomer. In our database we focuss just on tow processes (lend , transfer).

These tow processes (lend , transfer) lead us to determine our entities according to What objectes we need for a process to be done, also what attribute we need to describe every entity , The attribute dependes on What data we need to store about every entity in our database

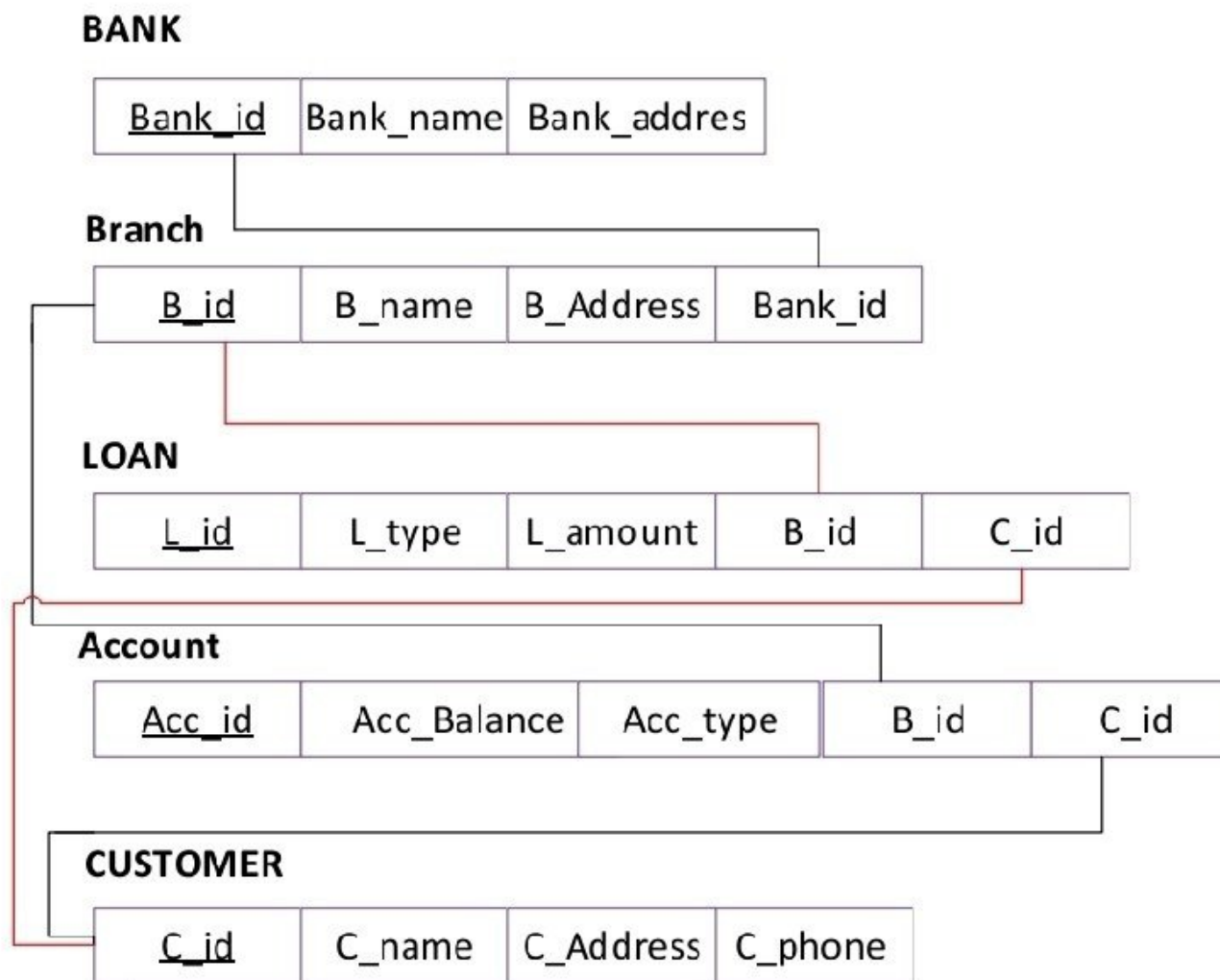
Phase I: Requirements collection and Analysis:

- 1- Banks: are identified by a Bank_name, Bank_id, Bank_address of main office,
The Primary key is Bank_id.
- 2- Branch: have coustomer and Branches are identified by B_id, B_name, B_address,
The Primary key is B_id.
- 3- Customers: are identified by C_id, C_name, C_phone number,C_address,
The Primary key is C_id.
- 4- Accounts: are identified by Acc_id, Acc_type, Acc_balance,
The Primary key is Acc_id.
- 5- Loans: are identified by L_id, L_type and, L_amount,
The Primary key is L_id.
- 6- Transications: are identified by T_id, T_type, from_id ,to_id , T_date, T_amount,
The Primary key is T_id

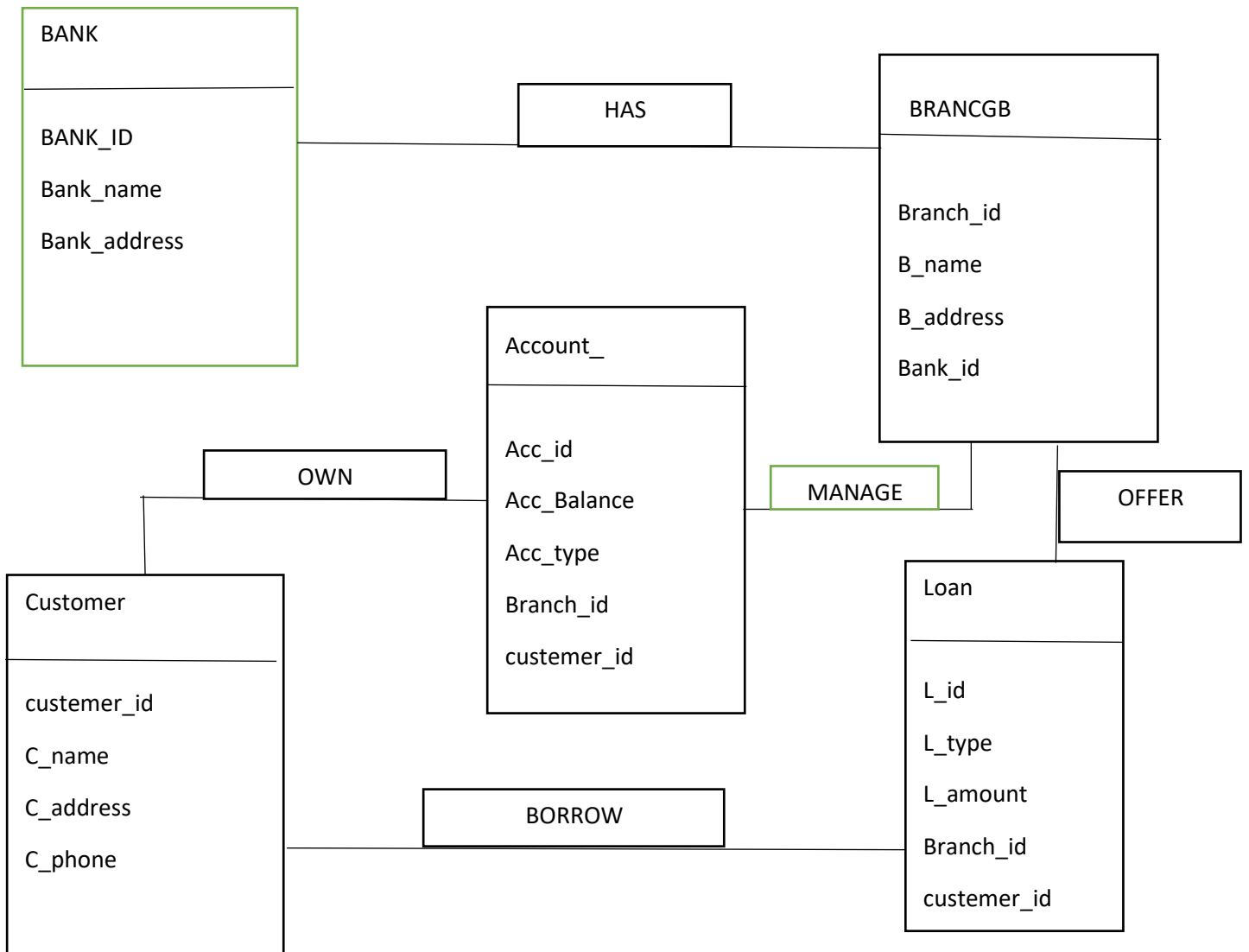
ER DIAGRAM :



PART 3 : Logical Model :



Phase IV: Physical schema design – Internal Schema:



PHASE V: QUERY FOR CREATING SUPERMARKET DATABASE:

```
create table Bank)
Bank_id number(5) Primary key,
Bank_name varchar2(20) not null,
Bank_address varchar2(40) not null
;(
insert into Bank values(1224,'RUAD','JDAH-Sudia');
insert into Bank values(2225,'AL ARBE','Reiady-Sudia');
insert into Bank values(3226,'RAGHE','JAZAN-Sudia');
select*from Bank;
```

```
9  select*from Bank ;|
```

BANK_ID	BANK_NAME	BANK_ADDRESS
1224	RUAD	JDAH-Sudia
2225	AL ARBE	Reiady-Sudia
3226	RAGHE	JAZAN-Sudia

```
create table BranchB(  
Branch_id number(5) primary key,  
B_name varchar2(20) not null,  
B_address varchar2(20) not null,  
Bank_id number(5)  
);  
  
insert into BranchB values(1112,'Alhakeika','DMAM-Sudia',2001);  
insert into BranchB values(2223,'Wathik','Reiady-Sudia',1002);  
insert into BranchB values(3334,'ZainB','ABHA-Sudia',3004);  
  
select*from BranchB;
```

```
10 select*from BranchB;
```

BRANCH_ID	B_NAME	B_ADDRESS	BANK_ID
1112	Alhakeika	DMAM-Sudia	2001
2223	Wathik	Reiady-Sudia	1002
3334	ZainB	ABHA-Sudia	3004

```
create table Customer(  
customer_id Number(5) primary key,  
C_name varchar2(20) not null,  
C_address varchar2(40) not null,  
C_phone varchar2(20) not null  
);  
  
insert into Customer values (1001,'hesham','JAZAN-KSA','+9960503344567');  
insert into Customer values (2003,'mohamad','JAZAN-KSA','+9660531755632');  
insert into Customer values (4003,'tamer','JAZAN-KSA','+9660522366478');  
insert into Customer values (4005,'ibrahem','JAZAN-KSA','+9660598766453');  
  
select*from Customer;
```

11 `select*from Customer;`

CUSTOMER_ID	C_NAME	C_ADDRESS	C_PHONE
1001	hesham	JAZAN-KSA	+9960503344567
2003	mohamad	JAZAN-KSA	+9660531755632
4003	tamer	JAZAN-KSA	+9660522366478
4005	ibrahem	JAZAN-KSA	+9660598766453

create table Loan)

L_id Number(5) primary key,

L_type varchar2(20) not null,

L_amount Float not null,

Branch_id Number, (°)

customer_id Number(5);((

insert into Loan values(1002,'Cash',2500,1112,2223);

insert into Loan values(2003,'Cash',12000,1113,11114);

insert into Loan values(3004,'Cash',15000,2114,3334);

select*from Loan

10 `select*from Loan;`

L_ID	L_TYPE	L_AMOUNT	BRANCH_ID	CUSTOMER_ID
1002	Cash	2500	1112	2223
2003	Cash	12000	1113	11114
3004	Cash	15000	2114	3334

```
create table Account(_
Acc_id Number(5) primary key,
Acc_Balance varchar2(20) not null,
Acc_type varchar2(40) not null,
Branch_id Number,(°)
custemer_id Number(5;((
insert into Account_ values(1002,20000,'aes',1001,2002);
insert into Account_ values(2003,50000,'aes',2223,3334);
insert into Account_ values(3004,20000,'aes',24445,1234);
select*from Account;_
```

```
10 select*from Account_;
```

ACC_ID	ACC_BALANCE	ACC_TYPE	BRANCH_ID	CUSTEMER_ID
1002	20000	aes	1001	2002
2003	50000	aes	2223	3334
3004	20000	aes	24445	1234

(Now We'll Perform all operations to all the SUPERMARKET attributes we had, Using PL/SQL programs)

1. **Declare** (in the Upper slides we've already declared every attribute and created tables with inserting it entities and data).
2. **Procedures.**
3. **Functions.**
4. **Cursors**
5. **Triggers.**

2. INSERT DATA TO SUPERMARKET DATABASE BY USING PROCEDURE

IN ORDER TO INSERT DATA INTO TABLE , WE FIRST NEED TO CREATE PROCEDURE AND PARAMETERS. THEN, WE CALL THAT PROCEDURE AND PUT THE RIGHT DATA INTO PARAMETER AS FOLLOWING:

- Bank\Input;

```
create or REPLACE PROCEDURE BANK_insert (
```

```
    B_Bank_id BANK .Bank_id%type,
```

```
    B_Bank_name BANK .Bank_name%type,
```

```
    B_Bank_address BANK .Bank_address%type  
)
```

```
AS
```

```
BEGIN
```

```
insert into BANK(Bank_id,Bank_name,Bank_address) values (B_Bank_id,B_Bank_name, B_Bank_address);  
commit;
```

```
DBMS_OUTPUT.PUT_LINE('record of ' || B_Bank_address || " " || " || 'inserted successfully' );  
exception
```

```
when others then
```

```
DBMS_OUTPUT.PUT_LINE('unexpected errors happened'|| ' '||sqlerrm);  
end;
```

```
1 create or REPLACE PROCEDURE BANK_insert (  
2     B_Bank_id BANK .Bank_id%type,  
3     B_Bank_name BANK .Bank_name%type,  
4     B_Bank_address BANK .Bank_address%type  
5 )  
6 AS  
7 BEGIN  
8 insert into BANK(Bank_id,Bank_name,Bank_address) values (B_Bank_id,B_Bank_name, B_Bank_address);  
9 commit;  
10 DBMS_OUTPUT.PUT_LINE('record of ' || B_Bank_address || ' ' || ' ' || 'inserted successfully' );  
11 exception  
12 when others then  
13 DBMS_OUTPUT.PUT_LINE('unexpected errors happened'|| ' '||sqlerrm);  
14 end;
```

Procedure created.

Call Procedure:

```
EXEC BANK_insert(77,'Mohammed','Sudia');
```

```
EXEC BANK_insert(57,'hesham','Sudia');
```

```
EXEC BANK_insert(68,'ibrahem','Sudia');
```

```
1 EXEC BANK_insert(77,'Mohammed','Sudia');  
2 EXEC BANK_insert(57,'hesham','Sudia');  
3 EXEC BANK_insert(68,'ibrahem','Sudia');
```

Statement processed.
record ofSudiainserted successfully

Statement processed.
record ofSudiainserted successfully

Statement processed.
record ofSudiainserted successfully

SELECT * FROM Bank;

77	Mohammed	Sudia
57	hesham	Sudia
68	ibrahem	Sudia

- BranchB\INPUT;

create or REPLACE PROCEDURE BranchB_INSERT(

B_Branch_id BranchB .Branch_id%type,

B_B_name BranchB .B_name%type,

B_B_address BranchB .B_address%type,

B_Bank_id BranchB .Bank_id%type

)

as

begin

INSERT INTO BranchB(Branch_id,B_name,B_address,Bank_id)

values(B_Branch_id,B_B_name, B_B_address,B_Bank_id);

commit;

DBMS_OUTPUT.PUT_LINE('record of' || B_Branch_id || ' '||'inserted successfully');

exception

when others then

DBMS_OUTPUT.PUT_LINE('unexpected errors happened'|| ' '||sqlerrm);

end;

```
1 create or REPLACE PROCEDURE BranchB_INSERT(
2     B_Branch_id BranchB .Branch_id%type,
3     B_B_name BranchB .B_name%type,
4     B_B_address BranchB .B_address%type,
5     B_Bank_id BranchB .Bank_id%type
6 )
7 as
8 begin
9     INSERT INTO BranchB(Branch_id,B_name,B_address,Bank_id) values( B_Branch_id,B_B_name, B_B_address,B_Bank_id);
10    commit;
11    DBMS_OUTPUT.PUT_LINE('record of' || B_Branch_id || ' '||'inserted successfully');
12 exception
13 when others then
14     DBMS_OUTPUT.PUT_LINE('unexpected errors happened'|| ' '||sqlerrm);
15 end;
```

Procedure created.

Call Procedure:

EXEC BranchB_insert(11,'THAMER','jazan',1);

EXEC BranchB_insert(22,'IBRAHEM','Abha',2);

EXEC BranchB_insert(33,'MOHAMMED','jazan',3);

```
1 EXEC BranchB_insert(11,'THAMER','jazan',1);
2 EXEC BranchB_insert(22,'IBRAHEM','Abha',2);
3 EXEC BranchB_insert(33,'MOHAMMED','jazan',3);
```

Statement processed.
record of11 inserted successfully

Statement processed.
record of22 inserted successfully

Statement processed.
record of33 inserted successfully

SELECT * FROM BranchB;

11	THAMER	jazan	1
22	IBRAHEM	Abha	2
33	MOHAMMED	jazan	3

• Customer\input;

create or REPLACE PROCEDURE Customer_insert(

C_customer_id Customer .customer_id%type,

C_C_name Customer .C_name%type,

C_C_address Customer .C_address%type,

C_C_phone Customer .C_phone%type
as

begin

INSERT INTO Customer (customer_id, C_name, C_address, C_phone)
values (C_customer_id, C_C_name, C_C_address, C_C_phone);

commit;

DBMS_OUTPUT.PUT_LINE('record of' || C_customer_id || " || 'inserted successfully');
13 exception

when others then

DBMS_OUTPUT.PUT_LINE('unexpected errors happened' || || sqlerrm);
end;

```
1 v create or REPLACE PROCEDURE Customer_insert(  
2   C_customer_id Customer .customer_id%type,  
3   C_C_name Customer .C_name%type,  
4   C_C_address Customer .C_address%type,  
5   C_C_phone Customer .C_phone%type  
6 )  
7 as  
8 begin  
9   INSERT INTO Customer (customer_id, C_name, C_address, C_phone)  
10  values (C_customer_id, C_C_name, C_C_address, C_C_phone);  
11  commit;  
12  DBMS_OUTPUT.PUT_LINE('record of' || C_customer_id || '' || 'inserted successfully');  
13 v exception  
14 when others then  
15   DBMS_OUTPUT.PUT_LINE('unexpected errors happened' || ' ' || sqlerrm);  
16 end;
```

Procedure created.

Call Procedure:

EXEC Customer_insert(12,' MOHAMMED', 'JAZAN','48557956');

EXEC Customer_insert(13,'IBRAHEM', 'JAZAN','468557956');

EXEC Customer_insert(14,'HESHAM', 'JAZAN','48557956');

```
1 EXEC Customer_insert(12,'MOHAMMED','JAZAN','48557956');  
2 EXEC Customer_insert(13,'IBRAHEM','JAZAN','468557956');  
3 EXEC Customer_insert(14,'HESHAM','JAZAN','48557956');
```

Statement processed.
record of12inserted successfully

Statement processed.
record of13inserted successfully

Statement processed.
record of14inserted successfully

SELECT * FROM Customer;

5 SELECT * FROM CUSTOMER;			
CUSTOMER_ID	C_NAME	C_ADDRESS	C_PHONE
12	MOHAMMED	JAZAN	48557956
13	IBRAHEM	JAZAN	468557956
14	HESHAM	JAZAN	48557956

- Loan\input;

create or REPLACE PROCEDURE Loan_insert(

L_L_id Loan .L_id%TYPE,

L_L_type Loan .L_type%TYPE,

L_L_amount Loan .L_amount%TYPE,

L_Branch_id Loan .Branch_id%TYPE

)

AS

begin

INSERT INTO Loan (L_id, L_type, L_amount Branch_id)

values (L_L_id, L_L_type, L_L_amount, L_Branch_id);

commit;

DBMS_OUTPUT.PUT_LINE('record of' || L_L_id || ' ' || 'inserted successfully');

exception

when others then

DBMS_OUTPUT.PUT_LINE('unexpected errors happened'|| ' ' || sqlerrm);

end;

```
1 v create or REPLACE PROCEDURE Loan_insert(  
2     L_L_id Loan .L_id%TYPE,  
3     L_L_type Loan .L_type%TYPE,  
4     L_L_amount Loan .L_amount%TYPE,  
5     L_Branch_id Loan .Branch_id%TYPE  
6 )  
7 AS  
8 begin  
9     INSERT INTO Loan (L_id, L_type, L_amount , Branch_id)  
10    values (L_L_id, L_L_type, L_L_amount, L_Branch_id);  
11    commit;  
12    DBMS_OUTPUT.PUT_LINE('record of' || L_L_id || ' ' || 'inserted successfully');  
13 v exception  
14 when others then  
15    DBMS_OUTPUT.PUT_LINE('unexpected errors happened'|| ' ' || sqlerrm);  
16 end;
```

Procedure created.

Call Procedure:

EXEC Loan_insert(100, 'Cash',25001,1);

EXEC Loan_insert(101, 'Cash', 120001,2);

EXEC Loan_insert(102, 'Cash',150002,3);

```

1 EXEC Loan_insert(100,'Cash',25001,1);
2 EXEC Loan_insert(101,'Cash',120001,2);
3 EXEC Loan_insert(102,'Cash',150002,3);

Statement processed.
record of100 inserted successfully

Statement processed.
record of101 inserted successfully

Statement processed.
record of102 inserted successfully

```

SELECT * FROM Loan;

100	Cash	25001	1	-
101	Cash	120001	2	-
102	Cash	150002	3	-

• Account\input;

create or REPLACE PROCEDURE Account__Insert(

A_Acc_id Account_ .Acc_id%TYPE,

A_Acc_Balance Account_ .Acc_Balance%TYPE,

A_Acc_type Account_ .Acc_type%TYPE,

A_Branch_id Account_ .Branch_id%TYPE,

A_customer_id Account_ .customer_id%TYPE

)

as

begin

INSERT INTO Account_ (Acc_id, Acc_Balance, Acc_type, Branch_id, customer_id)
values (A_Acc_id, A_Acc_Balance, A_Acc_type, A_Branch_id, A_customer_id);

commit;

DBMS_OUTPUT.PUT_LINE('record of' || A_Acc_id || ' ' || 'inserted successfully');

exception

when others then

DBMS_OUTPUT.PUT_LINE('unexpected errors happened'|| ' ' || sqlerrm);

end;

```

1 v create or REPLACE PROCEDURE Account__Insert(
2   A_Acc_id Account_ .Acc_id%TYPE,
3   A_Acc_Balance Account_ .Acc_Balance%TYPE,
4   A_Acc_type Account_ .Acc_type%TYPE,
5   A_Branch_id Account_ .Branch_id%TYPE,
6   A_customer_id Account_ .customer_id%TYPE
7 )
8 as
9 begin
10  INSERT INTO Account_ (Acc_id, Acc_Balance, Acc_type, Branch_id, customer_id)
11  values (A_Acc_id, A_Acc_Balance, A_Acc_type, A_Branch_id, A_customer_id);
12  commit;
13  DBMS_OUTPUT.PUT_LINE('record of' || A_Acc_id || ' ' || 'inserted successfully');
14 v exception
15 when others then
16  DBMS_OUTPUT.PUT_LINE('unexpected errors happened'|| ' ' || sqlerrm);
17 end;

```

Procedure created.

Call Procedure:

```
EXEC Account__Insert(11, 20000, 'Tawfier',11,1);  
EXEC Account__Insert(22,30000, 'Tawfier',22,2);  
EXEC Account__Insert (33, 40000, 'Tawfier',33,3);
```

```
1 EXEC Account__Insert(11, 20000, 'Tawfier',11,1);  
2 EXEC Account__Insert(22 ,30000, 'Tawfier',22,2);  
3 EXEC Account__Insert(33, 40000, 'Tawfier',33,3);|  
  
Statement processed.  
record of11 inserted successfully  
  
Statement processed.  
record of22 inserted successfully  
  
Statement processed.  
record of33 inserted successfully
```

SELECT * FROM Account_;

11	20000	Tawfier	11	1
22	30000	Tawfier	22	2
33	40000	Tawfier	33	3

3. SHOW DATA IN THE SUPERMARKET DATABASE BY USING FUNCTION:

FUNCTION IS A STANDALONE PL/SQL SUBPROGRAM. LIKE PL/SQL PROCEDURE, FUNCTIONS HAVE A UNIQUE NAME BY WHICH IT CAN BE REFERRED. THESE ARE STORED AS PL/SQL DATABASE OBJECTS.

- First Function:

CREATE OR REPLACE FUNCTION

functi

(name in VARCHAR2) RETURN VARCHAR2

IS

BEGIN

RETURN ('a suo:' || name);

END;

/

DECLARE

j VARCHAR2(250);

BEGIN

j:=functi('Project Students');

dbms_output.put_line(j);

END;

/

SELECT functi('ahmad , Ali, mayk')FROM DUAL;

```
1 v CREATE OR REPLACE FUNCTION
2 v functi
3 (name in VARCHAR2) RETURN VARCHAR2
4 IS
5 BEGIN
6 RETURN ('a suo:' || name);
7 END;
8 v /
```

Function created.

Statement processed.
a suo:Project Students

FUNCTI('AHMAD,ALI,MAYK')

a suo:ahmad , Ali, mayk

- Second Function:

```
CREATE OR REPLACE FUNCTION funct
(name in VARCHAR2) RETURN VARCHAR2
IS
BEGIN
RETURN ('We are pleased with your presence ' || name);
END;
/
DECLARE
j VARCHAR2(250);
BEGIN
j:=func('ABHA UNIVERSITY');
dbms_output.put_line(j);
END;
/
SELECT funct('IN OUR PROJECT')FROM DUAL;
```

```
1 v CREATE OR REPLACE FUNCTION funct
2   (name in VARCHAR2) RETURN VARCHAR2
3   IS
4   BEGIN
5   RETURN ('We are pleased with your presence ' || name);
6   END;
7 v /
8 DECLARE|
```

Function created.

Statement processed.

We are pleased with your presence ABHA UNIVERSITY

FUNCT('INOURLPROJECT')	
	We are pleased with your presence IN OUR PROJECT

4. INSERT DATA TO SUPERMARKET DATABASE BY

USING CURSOR:

IN SQL, A CURSOR MAKE IT POSSIBLE TO DEFINE A RESULT SET (A SET OF DATA ROWS) AND
PERFORM

COMPLEX LOGIC ON A ROW BY ROW BASIS, BY (INSERT AND DROP).

- BANK insert:

```
create table Bank(  
    Bank_id number(5) Primary key,  
    Bank_name varchar2(20) not null,  
    Bank_address varchar2(40) not null );  
insert into Bank values(1,'Alragihy','Gada-Sudia');  
insert into Bank values(2,'Enma','Reiady-Sudia');  
insert into Bank values(3,'Alahly','Gazan-Sudia');
```

```

1 v create table Bank(
2     Bank_id number(5) Primary key,
3     Bank_name varchar2(20) not null,
4     Bank_address varchar2(40) not null
5 );
6 insert into Bank values(1,'Alragihy','Gada-Sudia');
7 insert into Bank values(2,'Enma','Reiady-Sudia');
8 insert into Bank values(3,'Alahly','Gazan-Sudia');
9 select*from Bank;

```

BANK_ID	BANK_NAME	BANK_ADDRESS
1	Alragihy	Gada-Sudia
2	Enma	Reiady-Sudia
3	Alahly	Gazan-Sudia

Download CSV

- Cursor on BANK:

```

DECLARE
total_rows number(2);
BEGIN
UPDATE BANK SET BANK_ID=BANK_ID+ 500; IF sql%notfound THEN
dbms_output.put_line('no BANK selected'); ELSIF sql%found THEN
total_rows := sql%rowcount;
dbms_output.put_line(total_rows || BANK selected');
END IF;
END;

```

/

Select * from BANK;

DROP table BANK;

```
1  DECLARE
2  total_rows number(2);
3  BEGIN
4  UPDATE Bank SET BANK_ID=BANK_ID; IF sql%notfound THEN
5  dbms_output.put_line('no Bank selected'); ELSIF sql%found THEN
6  total_rows := sql%rowcount;
7  dbms_output.put_line(total_rows||'Bank selected');
8  END IF;
9  END;
10 Select * from Bank;
11 DROP table Bank
```

BANK_ID	BANK_NAME	BANK_ADDRESS
501	Alragihy	Gada-Sudia
502	Enma	Reiady-Sudia
503	Alahly	Gazan-Sudia

Download CSV

- Customer insert:

CREATE TABLE customer

(

Fname VARCHAR(15) NOT NULL,

Lname VARCHAR(15) NOT NULL,

CID VARCHAR(30),

Address VARCHAR(30),

ACCOUNTNO int)

```
INSERT INTO customer (Fname , Lname , CID , Address, ACCOUNTNO)
VALUES('IBRAHIM' , 'ALHARBI',1122,'jazan',212);
```

```
INSERT INTO customer (Fname , Lname , CID , Address, ACCOUNTNO)
VALUES('MOHMMED','ALJOMHAN',2231,'Abha',8975);
```

```
INSERT INTO customer (Fname , Lname , CID , Address, ACCOUNTNO)
VALUES('THMER','ERWY',13514,'jazan',124504);
```

Select * from customer;

```
1 ✓ CREATE TABLE customer
2 (
3   Fname VARCHAR(15) NOT NULL,
4   Lname VARCHAR(15) NOT NULL,
5   CID VARCHAR(30),
6   Address VARCHAR(30),
7   ACCOUNTNO int)
8 INSERT INTO customer (Fname , Lname , CID , Address, ACCOUNTNO)
9 VALUES('IBRAHIM' , 'ALHARBI',1122,'jazan',212);
10 ✓ INSERT INTO customer (Fname , Lname , CID , Address, ACCOUNTNO)
11 VALUES('MOHMMED','ALJOMHAN',2231,'Abha',8975);
12 ✓ INSERT INTO customer (Fname , Lname , CID , Address, ACCOUNTNO)
13 VALUES('THMER','ERWY',13514,'jazan',124504);
```

FNAME	LNAME	CID	ADDRESS	ACCOUNTNO
IBRAHIM	ALHARBI	1122	jazan	212
MOHMMED	ALJOMHAN	2231	Abha	8975
THMER	ERWY	13514	jazan	124504

- Cursor on Customer:

```
DECLARE
```

```
total_rows number(2);
```

```
BEGIN
```

```
UPDATE customer SET ACCOUNTNO=ACCOUNTNO + 1; IF sql%notfound THEN
```

```
dbms_output.put_line('no customer selected'); ELSIF sql%found THEN
```

```
total_rows := sql%rowcount;
```

```
dbms_output.put_line(total_rows || 'customer selected');
```

```
END IF;
```

```
END;
```

/

Select * from customer;

DROP table customer;

```
1 DECLARE
2   total_rows number(2);
3 BEGIN
4   UPDATE customer SET ACCOUNTNO=ACCOUNTNO + 1; IF sql%notfound THEN
5     dbms_output.put_line('no customer selected'); ELSIF sql%found THEN
6       total_rows := sql%rowcount;
7       dbms_output.put_line(total_rows||'customer selected');
8     END IF;
9   END;
10 Select * from customer;
11 DROP table customer;
```

FNAME	LNAME	CID	ADDRESS	ACCOUNTNO
IBRAHIM	ALHARBI	1122	jazan	213
MOHMMED	ALJOMHAN	2231	Abha	8976
THMER	ERWY	13514	jazan	124505

• Product insert:

create table product (

product_name Varchar(20),

P_ID Number(4) Primary key,

PClass_ID Number(6),

Price Number (3),

Expiration_date number(10),

Dept_number number(10),

Brand varchar2(10)

)

INSERT INTO product (product_name , P_ID , PClass_ID, Price ,

Expiration_date,Dept_number ,Brand)

VALUES('IBRAHIM',111,00001,12,2024,100,'pepsi');

INSERT INTO product (product_name , P_ID , PClass_ID, Price ,

Expiration_date,Dept_number ,Brand)

```
VALUES('ALI',222,00002,32,2028,100,'almarai');

INSERT INTO product (product_name , P_ID , PClass_ID, Price ,
                    Expiration_date,Dept_number ,Brand)
VALUES('HISHAM',333,00003,24,2023,300,'nutella');

INSERT INTO product (product_name , P_ID , PClass_ID, Price ,
                    Expiration_date,Dept_number ,Brand)
values('cornPuff',444,00004,55,2026,400,'cheetos');
```

```
1 v create table product (
2   product_name Varchar(20),
3   P_ID Number(4) Primary key,
4   PClass_ID Number(6),
5   Price Number (3),
6   Expiration_date number(10),
7   Dept_number number(10),
8   Brand varchar2(10)
9 )
```

```
10 INSERT INTO product (product_name , P_ID , PClass_ID, Price ,
11   Expiration_date,Dept_number ,Brand )
12   VALUES('IBRAHIM',111,00001,12,2024,100,'pepsi');
13 v INSERT INTO product (product_name , P_ID , PClass_ID, Price ,
14   Expiration_date,Dept_number ,Brand)
15   VALUES('ALI',222,00002,32,2028,100,'almarai');
16 v INSERT INTO product (product_name , P_ID , PClass_ID, Price ,
17   Expiration_date,Dept_number ,Brand)
18   VALUES('HISHAM',333,00003,24,2023,300,'nutella');
19 v INSERT INTO product (product_name , P_ID , PClass_ID, Price ,
20   Expiration_date,Dept_number ,Brand)
21   values('cornPuff',444,00004,55,2026,400,'cheetos');
22 select*from product;
```

PRODUCT_NAME	P_ID	PCLASS_ID	PRICE	EXPIRATION_DATE	DEPT_NUMBER	BRAND
IBRAHIM	111	1	12	2024	100	pepsi
ALI	222	2	32	2028	100	almarai
HISHAM	333	3	24	2023	300	nutella
cornPuff	444	4	55	2026	400	cheetos

- Cursor on Product:

```

DECLARE
total_rows number(5);
BEGIN
UPDATE product SET Price=Price + 5; IF sql%notfound THEN
dbms_output.put_line('no product selected'); ELSIF sql%found THEN
total_rows := sql%rowcount;
dbms_output.put_line(total_rows || 'product selected');
END IF;
END;/
Select * from product;
DROP table product;
```

```

1 v DECLARE
2   total_rows number(5);
3 v BEGIN
4 v UPDATE product SET Price=Price + 5; IF sql%notfound THEN
5 v dbms_output.put_line('no product selected'); ELSIF sql%found THEN
6   total_rows := sql%rowcount;
7   dbms_output.put_line(total_rows || 'product selected');
8   END IF;
9   END;
10 Select * from product;
11 DROP table product;
```

PRODUCT_NAME	P_ID	PCLASS_ID	PRICE	EXPIRATION_DATE	DEPT_NUMBER	BRAND
IBRAHIM	111	1	17	2024	100	pepsi
ALI	222	2	37	2028	100	almarai
HISHAM	333	3	29	2023	300	nutella
cornPuff	444	4	60	2026	400	cheetos

- Department insert:

```

CREATE TABLE Department
(
Deptname VARCHAR(15) NOT NULL,
Depttype VARCHAR(15) NOT NULL,
```



```

Depnumber number(30),
SSN int NOT NULL,
BranchID int
)

INSERT INTO Department (Deptname , Depttype , Depnumber , SSN , BranchID)
VALUES('fish','seafood',12,1,3);

INSERT INTO Department (Deptname, Depttype, Depnumber, SSN,BranchID)
VALUES('phone','Electronic',2,112,22);

INSERT INTO Department (Deptname, Depttype, Depnumber, SSN ,BranchID)
VALUES('coffee','Drinks',14,3,33);

Select * from Department ;

```

```

2 (
3 Deptname VARCHAR(15) NOT NULL,
4 Depttype VARCHAR(15) NOT NULL,
5 Depnumber number(30),
6 SSN int NOT NULL,
7 BranchID int
8 )
9 INSERT INTO Department (Deptname , Depttype , Depnumber , SSN , BranchID)
10 VALUES('fish','seafood',12,1,3);
11 INSERT INTO Department (Deptname, Depttype, Depnumber, SSN,BranchID)
12 VALUES('phone','Electronic',2,112,22);
13 INSERT INTO Department (Deptname, Depttype, Depnumber, SSN ,BranchID)
14 VALUES('coffee','Drinks',14,3,33);

```

DEPTNAME	DEPTTYPE	DEPNUMBER	SSN	BRANCHID
fish	seafood	12	1	3
phone	Electronic	2	112	22
coffee	Drinks	14	3	33

- Cursor on Department:

```

DECLARE
total_rows number(2);

BEGIN

UPDATE Department SET BranchID=BranchID + 1; IF sql%notfound THEN
dbms_output.put_line('no Department selected'); ELSIF sql%found THEN
total_rows := sql%rowcount;

```

```
dbms_output.put_line(total_rows || 'Department selected');
```

```
END IF;
```

```
END;
```

```
/
```

```
Select * from Department ;
```

```
DROP table Department ;
```

```
1 DECLARE
2   total_rows number(2);
3 BEGIN
4   UPDATE Department SET BranchID=BranchID + 1; IF sql%notfound THEN
5     dbms_output.put_line('no Department selected'); ELSIF sql%found THEN
6       total_rows := sql%rowcount;
7       dbms_output.put_line(total_rows || 'Department selected');
8     END IF;
9   END;
10  Select * from Department ;
11  DROP table Department ;
```

DEPTNAME	DEPTTYPE	DEPNUMBER	SSN	BRANCHID
fish	seafood	12	1	5
phone	Electronic	2	112	24
coffee	Drinks	14	3	35

• Branch insert:

```
1 Create table branch_
2 (
3   Branchname VARCHAR2(30),
4   Branch_ID number (5) PRIMARY KEY,
5   Address char(10),
6   SSN Number(10) NOT NULL,
7   C_ID Number(4)
8 );
9 INSERT INTO branch_ (Branchname , Branch_ID , Address, SSN ,C_ID )
10 VALUES('first',1001,'Riyadh',123,121);
11 INSERT INTO branch_ (Branchname , Branch_ID , Address, SSN ,C_ID)
12 VALUES('second',2002,'jeddah',124,102);
13 INSERT INTO branch_ (Branchname , Branch_ID , Address, SSN ,C_ID)
14 VALUES('fourth',4004,'abha',125,105);
```

BRANCHNAME	BRANCH_ID	ADDRESS	SSN	C_ID
first	1001	Riyadh	123	121
second	2002	jeddah	124	102
fourth	4004	abha	125	105

- Cursor on Branch:

```

DECLARE
total_rows number(5);

BEGIN
UPDATE branch_ SET Branch_ID =Branch_ID + 2; IF sql%notfound THEN
dbms_output.put_line('no branch_ selected'); ELSIF sql%found THEN
total_rows := sql%rowcount;
dbms_output.put_line(total_rows||'branch_ selected');
END IF;
END;
/
Select * from branch_;
DROP table branch_;

```

```

1 DECLARE
2   total_rows number(5);
3 BEGIN
4   UPDATE branch_ SET Branch_ID =Branch_ID + 2; IF sql%notfound THEN
5     dbms_output.put_line('no branch_ selected'); ELSIF sql%found THEN
6       total_rows := sql%rowcount;
7       dbms_output.put_line(total_rows||'branch_ selected');
8     END IF;
9   END;
10  Select * from branch_;
11  DROP table branch_;

```

BRANCHNAME	BRANCH_ID	ADDRESS	SSN	C_ID
first	1003	Riyadh	123	121
second	2004	jeddah	124	102
fourth	4006	abha	125	105

5. INSERT DATA TO SUPERMARKET DATABASE BY USING Trigger:

TRIGGERS PROVIDE A WAY OF EXECUTING PL/SQL CODE ON THE OCCURRENCE OF SPECIFIC DATABASE EVENTS

- First (Create tables):

create table Bank(Bank_id number(5) Primary key, Bank_name varchar2(20) not null, Bank_address varchar2(40) not null);

SQL Worksheet

```
1 v create table Bank(  
2   Bank_id number(5) Primary key,  
3   Bank_name varchar2(20) not null,  
4   Bank_address varchar2(40) not null  
5 );
```

Table created.

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create table BranchB(Branch_id number(5) primary key, B_name varchar2(20) not null, B_address varchar2(20) not null, Bank_id number(5) , CONSTRAINT FK_Bank_id FOREIGN KEY (Bank_id) REFERENCES Bank(Bank_id));

SQL Worksheet

```
1 v create table BranchB(  
2   Branch_id number(5) primary key,  
3   B_name varchar2(20) not null,  
4   B_address varchar2(20) not null,  
5   Bank_id number(5) ,  
6   CONSTRAINT FK_Bank_id FOREIGN KEY (Bank_id) REFERENCES  
7   Bank(Bank_id)  
8 );  
9
```

Table created.

- Second (Insert values in each table):

insert into Bank values(1,'Alragihy','Gada-Sudia'); insert into Bank values(2,'Enma','Reiady-Sudia'); insert into Bank values(3,'Alahly','Gazan-Sudia');

SQL Worksheet

```
1 v Begin  
2 insert into Bank values(1,'Alragihy','Gada-Sudia');  
3 insert into Bank values(2,'Enma','Reiady-Sudia');  
4 insert into Bank values(3,'Alahly','Gazan-Sudia');  
5 Commit;  
6 End;  
7 v /  
8 SELECT * FROM Bank;  
9 /  
10
```

BANK_ID	BANK_NAME	BANK_ADDRESS
1	Alragihy	Gada-Sudia
2	Enma	Reiady-Sudia
3	Alahly	Gazan-Sudia

insert into BranchB values(1,'Thamer','Gada-Sudia',2); insert into BranchB values(2,'hisham','Reiady-Sudia',1); insert into BranchB values(3,'Ibrahim','Gazan-Sudia',3);

SQL Worksheet

```
1 v Begin
2 insert into BranchB values(1,'Thamer','Gada-Sudia',2);
3 insert into BranchB values(2,'Hisham','Reiady-Sudia',1);
4 insert into BranchB values(3,'Ibrahim','Gazan-Sudia',3);
5 Commit;
6 End;
7 v /
8 SELECT * FROM BranchB;
9 /
```

Statement processed.

BRANCH_ID	B_NAME	B_ADDRESS	BANK_ID
1	Thamer	Gada-Sudia	2
2	Hisham	Reiady-Sudia	1
3	Ibrahim	Gazan-Sudia	3

- Third (Create a view for both tables):

Create view JOINED_Bank (Bank_name, Bank_id,B_name, Branch_ID) As Select Bank.Bank_name, Bank.Bank_id, BranchB.B_name, BranchB.Branch_ID From Bank, BranchB Where BrachB.Branch_ID=BranchB.Branch_ID / Select *from JOINED_Bank; /

SQL Worksheet

```
1 v Create view JOINED_Bank
2 ( Bank_name, Bank_id ,B_name, Branch_id)
3 As
4 Select Bank.Bank_name, Bank. Bank_id, BranchB.B_name, BranchB. Branch_id
5 From Bank, BranchB
6 Where BranchB.Branch_id=BranchB. Branch_id
7 /
8 Select *from JOINED_Bank;
9 /
```

View created.

BANK_NAME	BANK_ID	B_NAME	BRANCH_ID
Alragihy	1	Thamer	1
Alragihy	1	Hisham	2
Alragihy	1	Ibrahim	3

- Fourth (Create The Trigger):

Here we used trigger to chang the Branch name from "second" to "fourth" to the branch_ID (2022) from the joined tables;

```
CREATE TRIGGER JOINED_Bank INSTEAD OF UPDATE ON  
JOINED_Bank FOR EACH ROW BEGIN UPDATE branchB SET B_name =  
:new.B_name WHERE Branch_ID =:old.Branch_ID ; END; / Begin  
UPDATE JOINED_Bank SET B_name= 'Ali' WHERE Branch_ID = 2;  
COMMIT; END; / SELECT * FROM JOINED_Bank;
```

SQL Worksheet

```
10  
11 CREATE TRIGGER JOINED_Bank  
12 INSTEAD OF UPDATE ON JOINED_Bank  
13 FOR EACH ROW  
14 BEGIN  
15 UPDATE BranchB SET B_name := new.B_name WHERE Branch_id :=old.Branch_id  
16 ;  
17 END;  
18 /  
19 Begin  
20 UPDATE BranchB SET B_name = 'Ali' WHERE Branch_id = 2;  
21 COMMIT;  
22 END;  
23 /  
24 SELECT * FROM JOINED_Bank;
```

BANK_NAME	BANK_ID	B_NAME	BRANCH_ID
Alragihy	1	Thamer	1
Alragihy	1	Ali	2
Alragihy	1	Ibrahim	3
Enma	2	Thamer	1